

Benchmarking Vision Architectures in Federated Learning under Data Heterogeneity for Medical Image Classification

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Abstract—Federated Learning (FL) enables collaborative model training across decentralized clinical institutions without sharing raw patient data, effectively complying with strict privacy regulations. However, FL faces significant challenges when dealing with heterogeneous (non-IID) data distributions typical in real-world medical settings. While Convolutional Neural Networks (CNNs) have been the standard, recent advances in Vision Transformers (ViTs) and State-Space Models (e.g., Vision Mamba) offer promising alternatives due to their global context modeling capabilities. This proposal presents a comprehensive comparative study of modern vision architectures—specifically EfficientNet, DeiT, and Vision Mamba—under FL settings with varying degrees of data heterogeneity. Despite the initial hypothesis that global context models would strictly outperform CNNs, preliminary evidence suggests performance varies significantly depending on the dataset and FL configuration. Therefore, this research aims to systematically benchmark these architectures on medical datasets (e.g., Brain Tumor MRI), comparing standard aggregation methods (FedAvg) against heterogeneity-robust algorithms (FedProx). The study will evaluate performance consistency, client-level fairness, and convergence stability over 100 communication rounds with multiple random seeds. The findings will provide crucial empirical insights into architectural robustness, guiding the deployment of reliable diagnostic models in federated healthcare environments.

Index Terms—Federated Learning, Data Heterogeneity, Vision Transformers, Convolutional Neural Networks, State-Space Models, Medical Imaging

I. INTRODUCTION

Privacy laws such as HIPAA and GDPR strictly limit data sharing in healthcare. Federated Learning (FL) addresses this by training algorithms locally on hospital data and aggregating the model updates on a central server. A critical challenge in FL is the inherent heterogeneity of medical data across institutions, featuring both feature and label skew. The significant performance drop of FL under these non-IID conditions necessitates robust model architectures.

While much research has focused on novel aggregation algorithms to mitigate heterogeneity, there is a lack of comprehensive benchmarks evaluating how modern vision architectures—such as Vision Transformers (DeiT) and State-Space Models (Vision Mamba)—compare against established Con-

volutional Neural Networks (CNNs) under non-IID medical scenarios. This research aims to systematically evaluate and compare the performance, robustness, and stability of CNNs, ViTs, and State-Space Models in FL environments with data heterogeneity for medical image classification. The expected contributions include a robust benchmark methodology for medical FL, a deep analysis of architectural behaviors under non-IID settings, and practical guidelines for model selection in clinical FL deployments.

II. RELATED WORK

Recent literature highlights the interplay between architecture and FL performance. Himeur et al. [1] investigated FL for computer vision, indicating that model architecture significantly impacts performance under non-IID data distributions, and Transformers can outperform CNNs under high heterogeneity. In the medical domain, Yan et al. [2] explored label-efficient self-supervised FL for tackling heterogeneity, bridging the gap between missing labels and non-IID data. Albalawi et al. [3] and Zhou et al. [4] focused on brain tumor classification, demonstrating how integrating FL with transfer learning and dedicated privacy-preserving mechanisms can enhance diagnostic reliability. Furthermore, recent studies outside of FL show models like Mamba remaining highly competitive in medical tasks, but not strictly dominant over strong CNNs like EfficientNet [5].

A major limitation of the current literature is that it predominantly focuses on novel aggregation algorithms or evaluates models in isolated IID settings rather than deeply analyzing the architectural resilience to data heterogeneity, particularly overlooking the latest sequence models like Vision Mamba in a controlled FL context.

III. RESEARCH PROBLEM

Definition: How do different vision architectures (CNNs, Transformers, State-Space Models) behave and compare in Federated Learning settings characterized by high data heterogeneity in medical image classification?

Research Questions:

- 1) Why do CNNs outperform or underperform fully attention-based models (DeiT) in specific medical datasets and FL setups?
- 2) Why do State-Space Models (Vision Mamba) struggle with convergence and stability in certain non-IID FL configurations compared to CNNs?
- 3) How does the type and degree of data heterogeneity affect the client-level fairness and global performance disparity of each architecture?

Hypothesis: DeiT and Vision Mamba can outperform CNNs in some scenarios characterized by high heterogeneity and when modeling global context is critical. However, they do not do so consistently across all datasets and FL configurations; CNN architectures (such as EfficientNet) remain highly competitive and tend to exhibit more stable convergence in early training stages.

IV. METHODOLOGY

The study will employ a Deep Learning experimental framework simulating cross-silo FL scenarios. The primary dataset will be the Brain Tumor MRI Dataset, with controlled non-IID splits using Dirichlet distributions for label skew. The selected models for the comparative study are EfficientNet-B0 (CNN), DeiT-tiny (Vision Transformer), and Vim-tiny (State-Space Model).

The FL pipeline entails standardized local training parameters across clients (optimizer, learning rate, batch size, local epochs) to isolate architectural effects. Two primary aggregation methods will be tested: FedAvg (baseline) and FedProx (heterogeneity-robust). The models will be evaluated using standard metrics (Accuracy, Precision, Recall, F1-Score), but specifically analyzed via mean and standard deviation over at least three independent random seeds. The validation strategy guarantees experimental rigor by setting a minimum of 100 communication rounds to capture long-term convergence patterns. Furthermore, the evaluation will include an analysis of performance per client to address variability and identify if specific clients suffer disproportionately (fairness).

V. RESEARCH PLAN

The project will be executed in the following tentative phases:

- **Phase 1 (Weeks 1-3):** Literature review refinement and experimental setup standardization, ensuring consistent seeds, hyperparameters, and partition strategies.
- **Phase 2 (Weeks 4-6):** Execution of the scaled experimental space (100 rounds, 3 seeds) for EfficientNet, DeiT, and Vim using FedAvg and FedProx on the medical dataset.
- **Phase 3 (Weeks 7-9):** Deep quantitative and qualitative analysis of results, focusing on client-level variability, architectural vulnerabilities, and convergence stability.
- **Phase 4 (Weeks 10-12):** Drafting the discussion section, contrasting findings with existing literature, and deducing theoretical insights for model behavior.
- **Phase 5 (Weeks 13-16):** Final review, document formatting for publication, and thesis defense preparation.

VI. EXPECTED IMPACT

Scientific: The study provides much-needed empirical evidence and theoretical discussion on how distinct vision architectures process heterogeneous decentralized data, contributing to both FL algorithm design and neural architecture research.

Applied: Offers pragmatic guidelines for technical teams in hospitals and ML engineers regarding which model architectures to deploy based on the specific network and data distribution characteristics of their federated consortium.

Ethical: By critically addressing client-level fairness, the research ensures that models do not severely underperform for institutions with minority data distributions, thus mitigating bias in decentralized diagnostic tools.

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